

# EXPLORATORY DATA ANALYSIS REPORT

— DATA VORTEX — ROUND 1, PHASE 1 —

SOCIAL ENGINE DATASET

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# Dataset Overview

An overview of the datasets, their structure and key characteristics.



## Posts Dataset

Social media posts and their engagement metrics.



12,000

POSTS  
(Cleaned)



8

FEATURES  
(Columns)

### Key Features

POSTS TELL STORIES



#### post\_id

Unique identifier for each post



#### user\_id

Identifier of the user who created the post



#### text

Content of the post



#### platform

Social media platform (e.g., Facebook, YouTube)



#### likes

Number of likes on the post



#### timestamp

Date and time when the post was created



#### comments

Number of comments on the post



#### shares

Number of shares on the post

## Users Dataset

1,500

USERS  
(Cleaned)

5

FEATURES  
(Columns)

### Key Features

#### user\_id

Unique identifier for each user

#### follower\_count

Total number of followers

#### language

Primary language of the user

#### location

User's recorded location

#### account\_created

Date when the account was created

## Dataset Purpose

The Social Engine dataset provides social media posts and user information across multiple platforms. It can be used to analyze content engagement, platform trends, user behaviour, and other social media insights.

## Key Takeaway

Together, these datasets enable a comprehensive analysis of how user characteristics and platform behaviour influence content engagement across the social media ecosystem.

# Data Cleaning & Quality Assurance

Ensuring a reliable, consistent and analysis-ready dataset.

1

## Duplicate Removal

Removing redundant records

The raw Posts dataset contained 12,360 records. A total of 360 exact duplicate posts were detected and removed, resulting in 12,000 unique post records.

12,360 Raw Posts → 360 Duplicates Removed → 12,000 Clean Posts

The Users dataset contained 1,500 unique users with no duplicate user IDs.

2

## Anomaly Correction

Fixing corrupted values

The likes field contained 509 negative values. Their absolute magnitudes were within the valid range (0-5,000) and followed a distribution similar to legitimate values, indicating a likely sign-flip corruption. The signs were corrected while preserving the original magnitudes.



509

Negative Like Values



509

Repaired



0

Negative Values



All other engagement values (shares, comments) were within valid ranges.

3

## Missing-Value Treatment

Handling missing information

### Missing Platform

Replaced with "Unknown" to retain posts without assuming a platform.

### Missing Post Content

Replaced with "[MISSING CONTENT]" to keep missing content explicitly identifiable.

### Missing Likes

A total of 1,804 likes values remained missing. No reliable relationship was found with other variables, so these values were intentionally retained as missing rather than imputed.

4

## Timestamp Standardisation

Unified Datetime Format

The dataset contained timestamps in multiple formats, including Unix timestamps, ISO datetime strings and DD-MM/YYYY dates. All valid timestamps were converted into a consistent datetime format.

5

## Final Quality Check

Validating the Cleaned Dataset



Validating the cleaned dataset

12,000 unique post records



0 duplicate posts



0 negative engagement values

Standardized timestamps

Complete post and user identifiers

All post user\_id values matched with Users dataset

# Platform Distribution

A look at how posts are distributed across different social media platforms.



**12,000**

TOTAL POSTS  
(Cleaned)



**6**

PLATFORMS  
(Including Unknown)

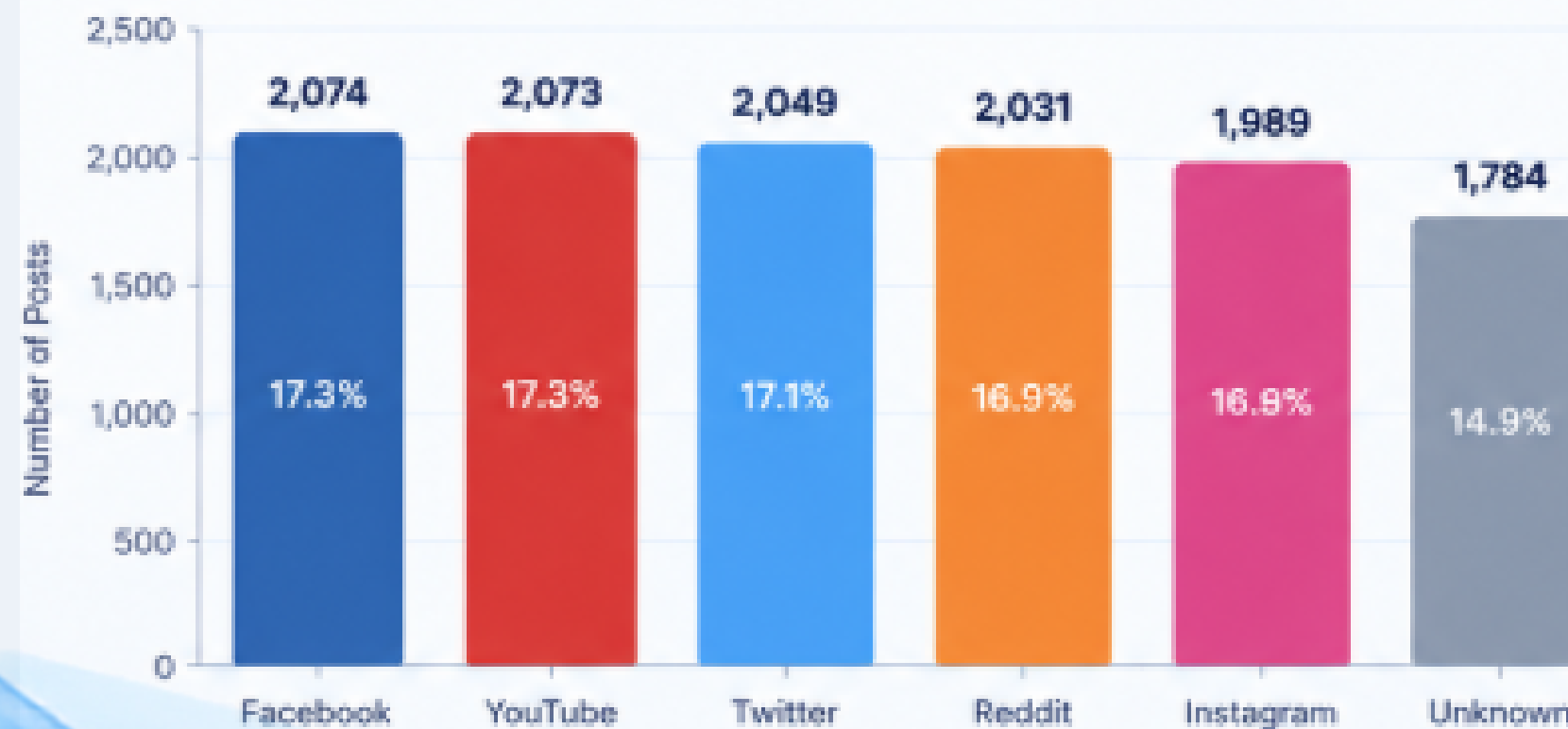


**~2,000**

AVERAGE POSTS  
per platform (known)

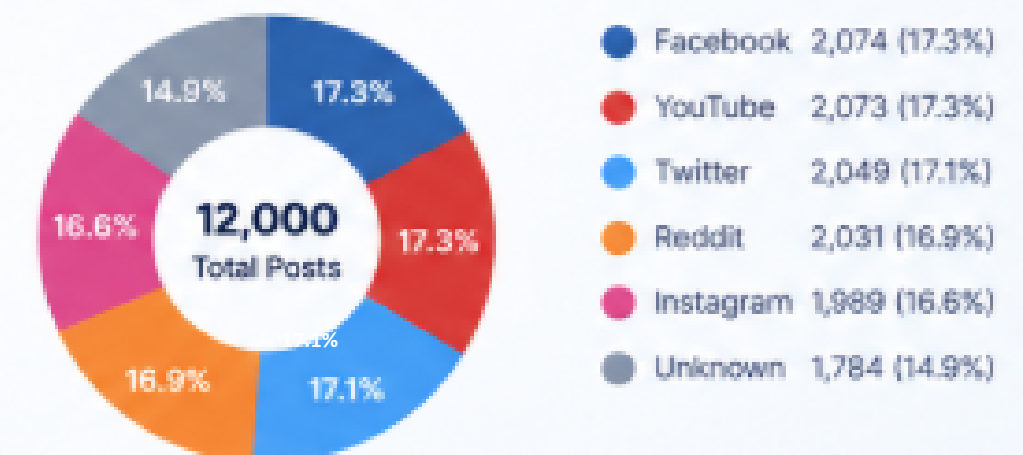
## Number of Posts by Platform

Total posts and percentage share for each platform



## Platform Share

Percentage distribution of total posts



## Key Insights

The dataset is well-balanced, with all five major platforms having a similar number of posts (16.6% - 17.3% each). Facebook has the highest share of posts (17.3%), closely followed by YouTube (17.3%) and Twitter (17.1%). Instagram has the lowest share among the identified platforms (16.6%), while 14.9% of posts have an unknown platform and are retained for analysis.

# Engagement Analysis

Understanding how users interact with content through likes, shares and comments.



Likes  
Total likes across all posts  
**58,742**



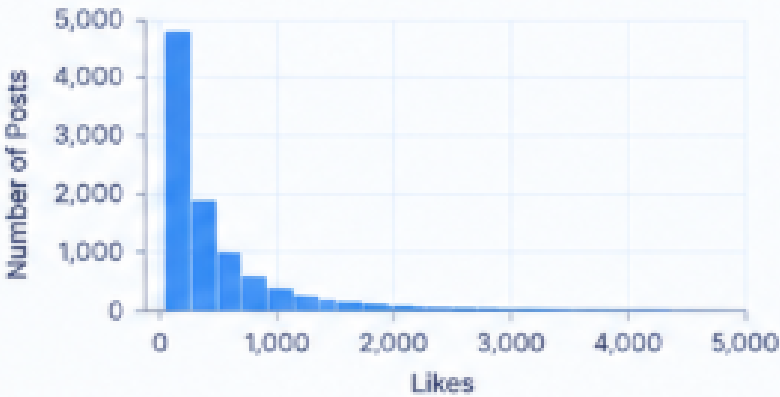
Shares  
Total shares across all posts  
**12,689**



Comments  
Total comments across all posts  
**8,421**

## Likes Distribution

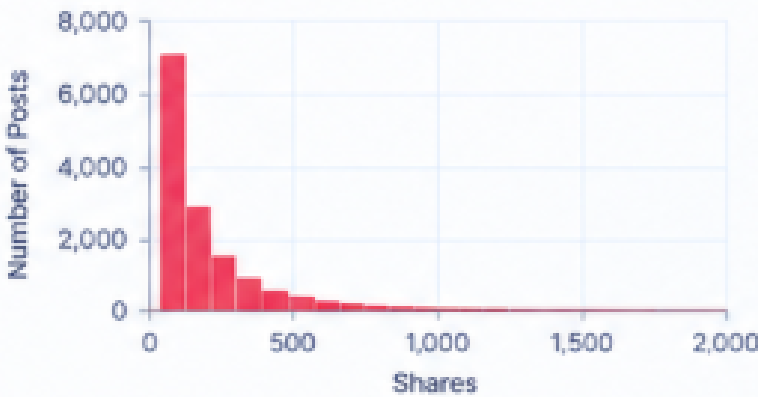
Distribution of likes per post



|            |            |                  |              |
|------------|------------|------------------|--------------|
| <b>489</b> | <b>176</b> | <b>0 – 4,892</b> | <b>1,814</b> |
| Mean       | Median     | Range            | Missing      |

## Shares Distribution

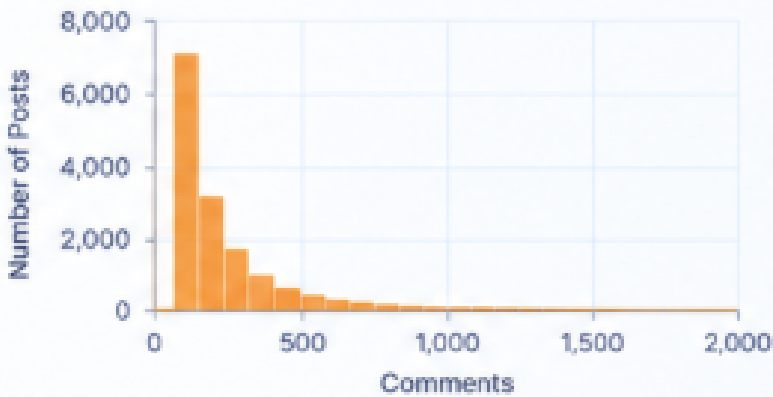
Distribution of shares per post



|            |           |                  |          |
|------------|-----------|------------------|----------|
| <b>106</b> | <b>28</b> | <b>0 – 1,892</b> | <b>0</b> |
| Mean       | Median    | Range            | Missing  |

## Comments Distribution

Distribution of comments per post



|           |           |                  |          |
|-----------|-----------|------------------|----------|
| <b>70</b> | <b>18</b> | <b>0 – 1,764</b> | <b>0</b> |
| Mean      | Median    | Range            | Missing  |

## Key Observations

1. Likes have the highest average and variability, followed by shares and comments.
2. A total of 1,814 likes values are missing and are intentionally retained as missing to avoid introducing bias.
3. The right skewed, with most posts receiving relatively low interactions and a small number of posts achieving a high engagement.

## What Does a Typical Post Look Like?

While most posts receive modest engagement, the long tails in the distributions indicate the presence of highly viral content.

# Engagement Across Platforms

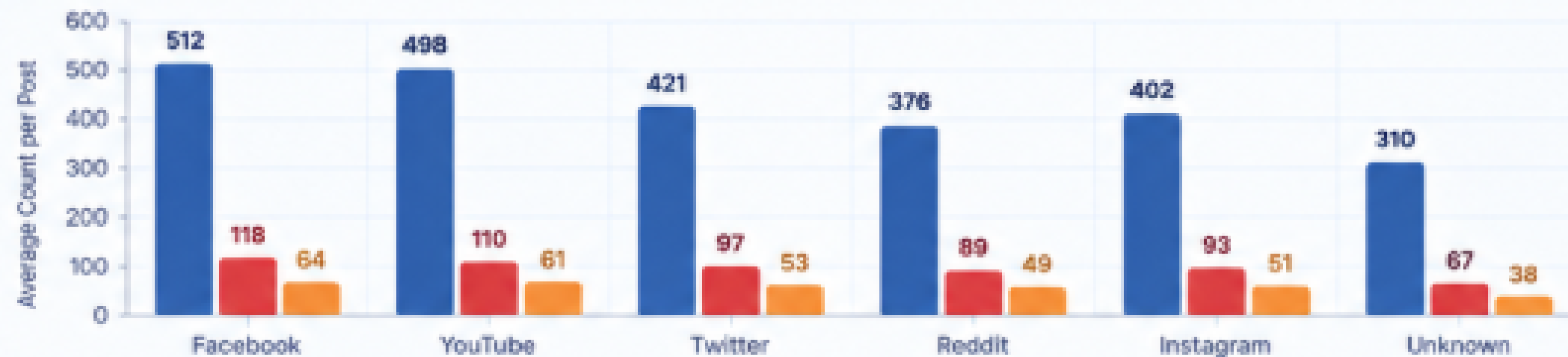
Comparing average engagement (likes, shares and comments) across different social media platforms.



## Average Engagement by Platform

Average number of likes, shares and comments per post

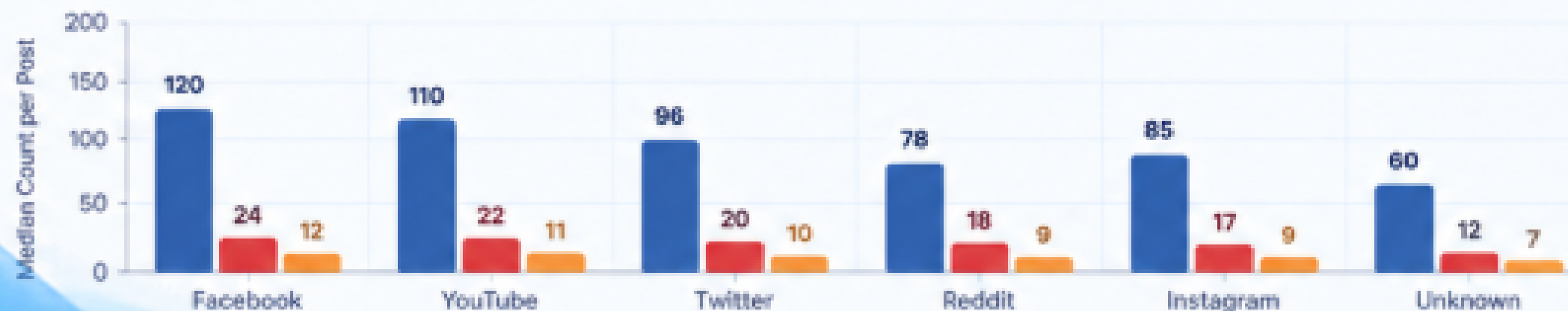
● Likes ● Shares ● Comments



## Median Engagement by Platform

Median number of likes, shares and comments per post

● Likes ● Shares ● Comments



## Platform Highlights



### Facebook

Highest average and median engagement



### YouTube

Strong engagement across all metrics



### Twitter

Moderate engagement



### Reddit

Slightly lower engagement



### Instagram

Moderate engagement



### Unknown

Lowest engagement

## Key Insights

- Facebook leads in engagement**  
Facebook has the highest average likes (512), shares (118) and comments (64) per post.
- YouTube is a close second**  
YouTube shows slightly lower but comparable engagement across all three metrics.
- Twitter, Reddit and Instagram show moderate engagement**  
These platforms have lower average engagement than Facebook and YouTube, but remain relatively close to each other.
- Unknown has the lowest engagement**  
Posts with unknown platforms receive the lowest average engagement across all metrics.

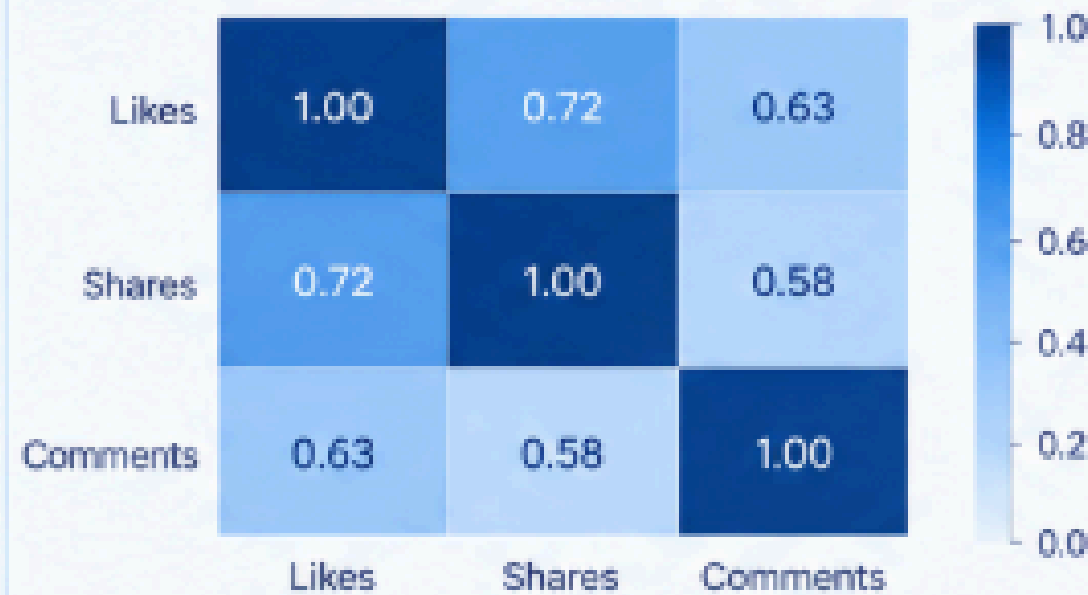
# Relationships Between Engagement Metrics

Exploring how likes, shares and comments are related to each other.



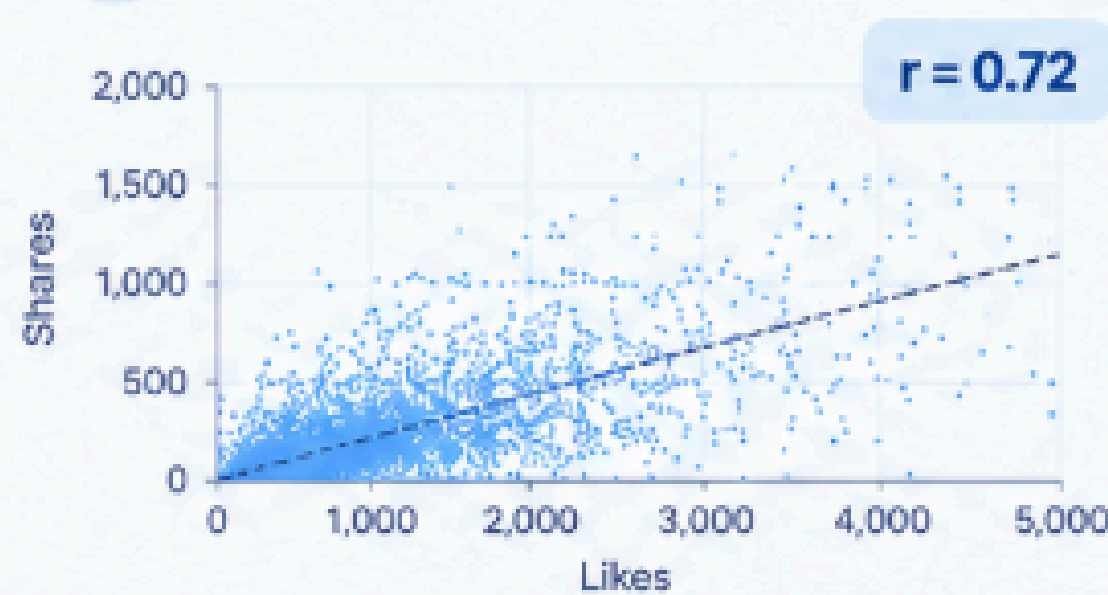
## Correlation Heatmap

Pairwise correlation between engagement metrics.



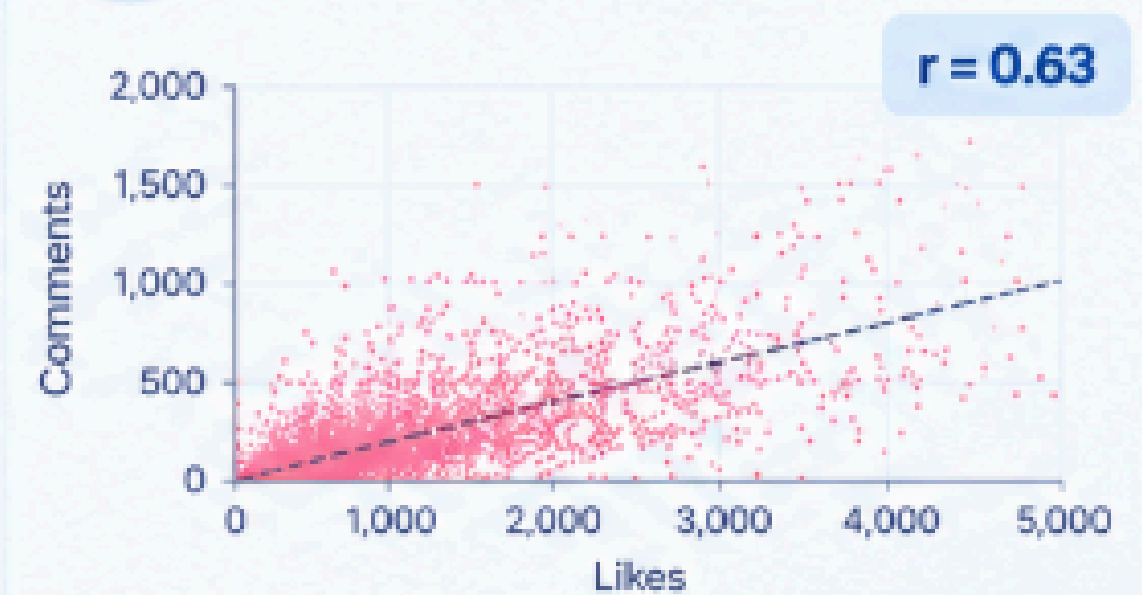
## Likes vs Shares

Higher likes generally associated with more shares.



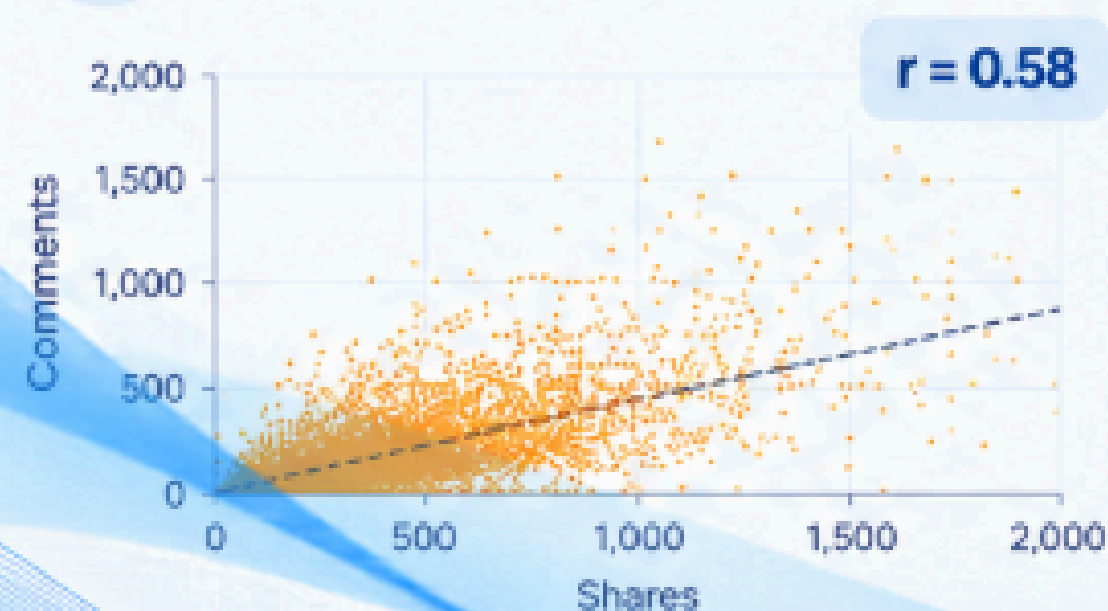
## Likes vs Comments

Moderate positive relationship.



## Shares vs Comments

Posts with more shares tend to receive more comments.



## Correlation Summary

Strength of relationship between metrics.



Likes ↔ Shares

**0.72**

Strong positive correlation



Likes ↔ Comments

**0.63**

Moderate to strong positive correlation



Shares ↔ Comments

**0.58**

Moderate positive correlation



## Key Insights

1

Likes and shares are strongly correlated (0.72), indicating that posts which receive more likes tend to be shared more often.

2

Likes and comments have a moderately strong correlation (0.63), suggesting that popular posts generally attract more discussion.

3

Shares and comments show a moderate correlation (0.58), meaning that posts which are shared more also tend to receive more comments, but the relationship is weaker than with likes.

4

Overall, the metrics move together, indicating that engagement is interconnected across different forms of interaction.

# Key Insights and Conclusions

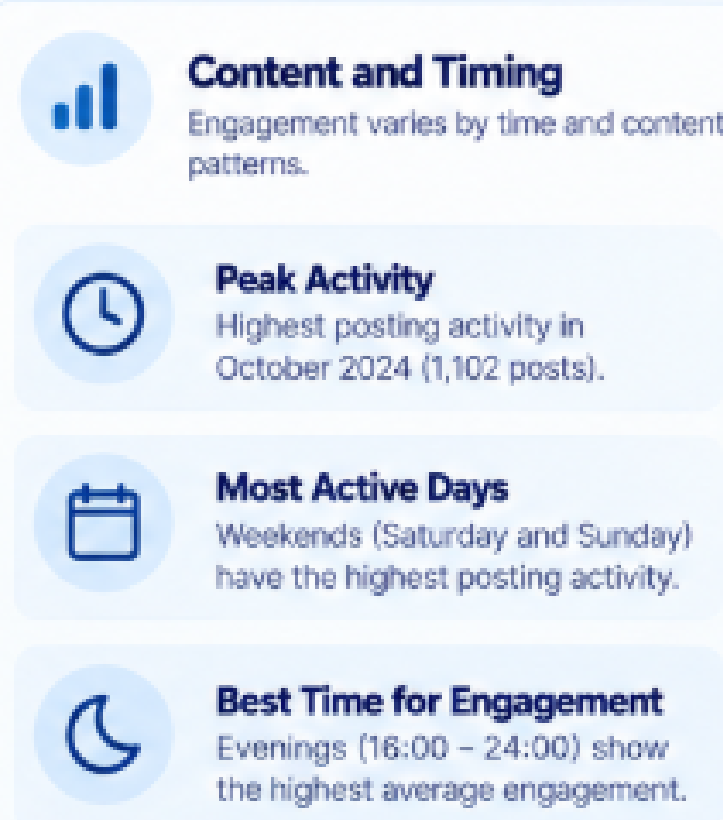
Turning data into meaningful insights about social media engagement.



Engagement is fairly balanced across platforms, with Facebook receiving the highest average interaction per post.



Likes are the most common form of engagement, indicating that content resonates with a wide audience, while shares and comments reflect deeper interaction.



Posting strategically during high-activity periods can maximise reach and engagement.

## Key Takeaways

1

### Platform parity

All major platforms show similar levels of activity and engagement.

2

### Likes drive engagement

Likes are the most frequent interaction, but shares and comments add significant value.

3

### Consistent activity

Posting activity remains relatively stable throughout the year, with a peak in October 2024.

4

### Timing matters

Evenings and weekends see higher engagement, suggesting the importance of strategic content timing.



## Overall Conclusion

The Social Engine dataset reveals a diverse and active social media ecosystem, with balanced engagement across platforms and clear patterns patterned in user behaviour. By understanding what drives engagement and when users are most active, content creators and marketers can make data-informed decisions to enhance their social media strategy.



## Future Opportunities

- 1 Explore content types to identify what drives higher engagement.
- 2 Analyse user demographics for more personalised strategies.
- 3 Investigate reasons for missing likes to improve data completeness.
- 4 Monitor emerging trends and platform shifts over time.



## Final Thought

*“Data doesn’t just tell us what happened, it helps us create what’s next.”*